# PROJECT DOCUMENTATION

Title – HID Based Robotic Eyes Using ESP32


## Abstract

This project presents the design and development of a Human Interface Device (HID) based robotic eye system controlled through keyboard inputs from a computer. The system enables realistic eye movements such as opening, closing, blinking, and eyeball movement in left, right, up, and down directions. Python is used to detect human keyboard interaction and transmit commands via serial communication. An ESP32 microcontroller receives these commands and controls multiple servo motors accordingly. The project demonstrates effective integration of software control, embedded systems, and mechanical design for real-time human–robot interaction.


## Introduction

Human–robot interaction is an important aspect of humanoid robotics and animatronics. Among various facial features, eyes play a crucial role in conveying expressions and attention. This project focuses on developing a robotic eye system that can be directly controlled by a human using a keyboard. Unlike autonomous eye-tracking systems, this approach provides full manual control with minimal hardware and computational complexity. The use of ESP32 and Python makes the system cost-effective, flexible, and suitable for educational and research applications.

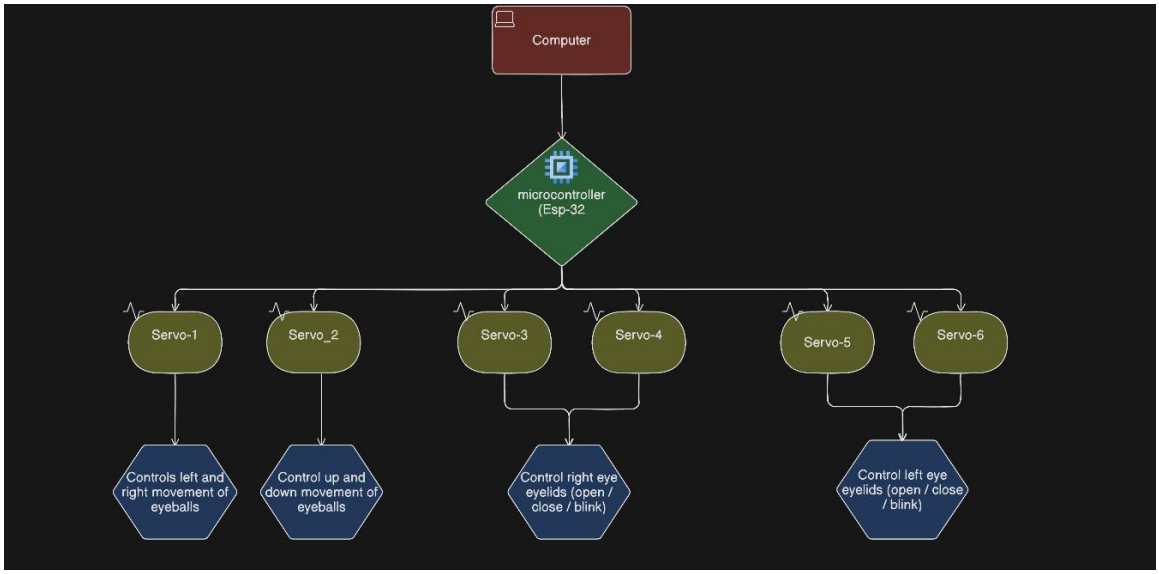
## Materials Used

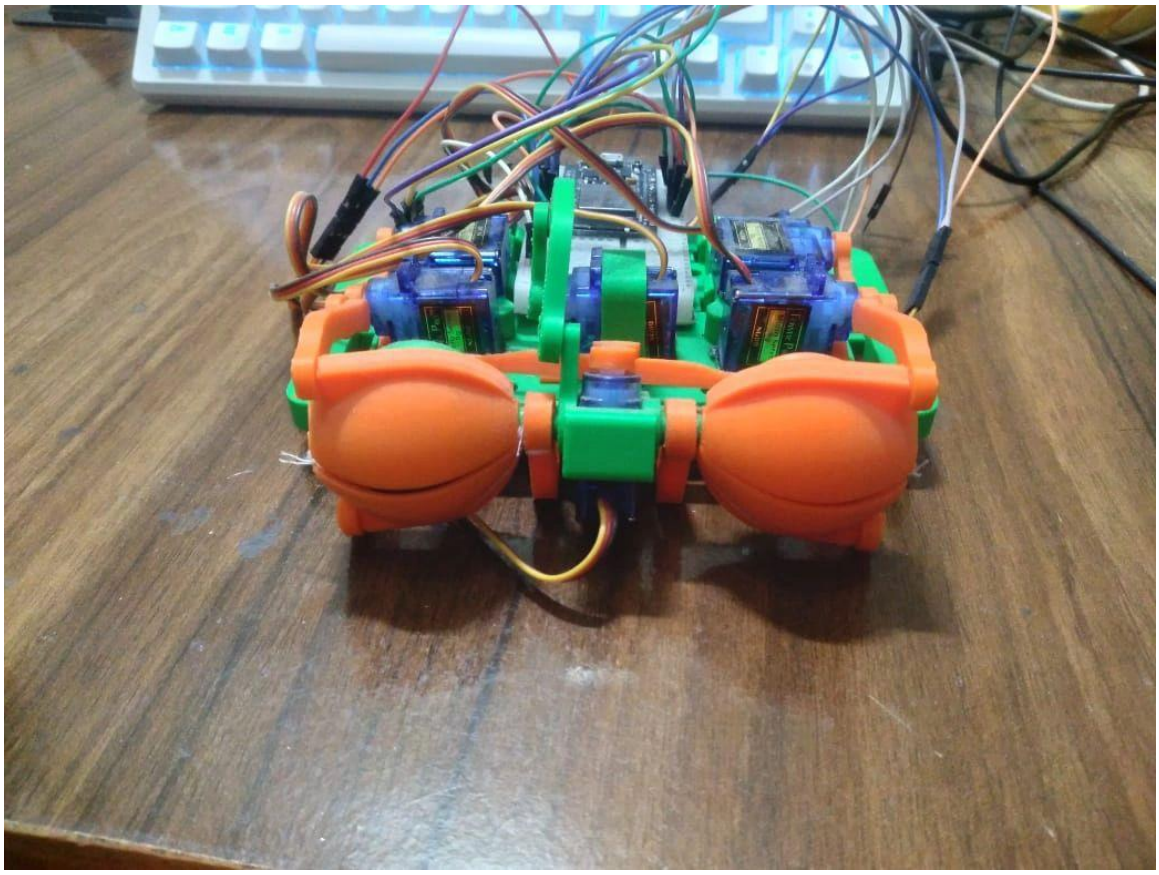
- ESP32 Microcontroller
- Servo Motors (SG90/Micro Servos)
- 3D Printed Robotic Eye Frame
- Jumper Wires
- External Power Supply
- Computer System
- USB Cable
- Arduino IDE
- Python (PySerial Library)

## Design, Schematic and Block Diagram

The system design consists of two main sections: the computer control unit and the embedded control unit. The computer detects keyboard inputs using Python and sends

corresponding commands through serial communication. The ESP32 receives these commands and generates PWM signals to control servo motors responsible for eye movement and eyelid motion.



## Implementation

The implementation process includes mechanical assembly, electronic interfacing, programming, and system integration.

1. Mechanical Assembly: The robotic eyes are mounted on a 3D-printed frame with servo motors responsible for horizontal, vertical, and eyelid movements.

2. Electronic Circuit: Servo motors are connected to the ESP32 PWM pins. An external power supply is used to provide adequate current for servo operation.

3. Programming: Python detects keyboard inputs and sends commands via serial communication. The ESP32 firmware reads these commands and moves the servos accordingly.

4. Calibration: Servo angles are calibrated to ensure smooth and realistic eye movement.

## Results and Analysis

The robotic eye system successfully responds to keyboard commands in real time. Eye movements such as blinking, opening, closing, and directional motion were achieved with minimal delay. The system showed stable performance and accurate servo positioning during repeated testing.

## Discussion and Limitations

While the system performs effectively, it has certain limitations. The setup relies on wired serial communication, limiting mobility. Servo motors provide limited precision, and the system does not include autonomous or vision-based control.

## Conclusion

The HID based robotic eyes project successfully demonstrates real-time human-controlled eye movements using Python and ESP32. The project highlights the practical integration of software and hardware to achieve expressive robotic behavior and serves as a strong foundation for future humanoid robotics applications.

## Future Scope

- Wireless control using Bluetooth or Wi-Fi
- AI-based eye tracking
- Emotion simulation
- Integration with facial expression systems
- Mobile or web-based control interface

## References

1. ESP32 Technical Documentation
2. Python PySerial Documentation
3. Arduino Servo Library

## Acknowledgement

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